

AUTOCAD AND SOLIDWORKS

FOR

MECHANICAL ENGINEERS

A CERTIFICATE COURSE CONDUCTED

BY



THE SURE Trust

Skill Upgradation for Rural-youth Empowerment – Trust

<https://suretrustforruralityouth.com/>

Course Training Attended

By

J.Vivek

Keshav kumar

Raghuveer sharma

Thejas K C

DECLARATION

We hereby declare that the project work titled “**Origami-Based Collapsible Roof System**” is a bonafide work carried out by us in partial fulfillment of the requirements for this internship.

This work has been carried out under the guidance of **Prof. Radhakumari Ma’am** and **Mr. Mairala Chinnarao**. We further declare that this report has not been submitted previously to any other organization.

J. Vivek

Keshav Kumar

Raghuveer Sharma

Thejas K C

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to **Sure Trust Organization** for providing us with the opportunity and platform to pursue this internship and carry out our project work on “Origami-Based Collapsible Roof System.” The exposure and learning experience gained through this initiative have been invaluable in enhancing our technical knowledge and practical skills.

We are deeply thankful to our faculty guide, **Prof. Radhakumari Ma’am**, for her constant encouragement, valuable suggestions, and continuous support throughout the course of this project. Her guidance has played a crucial role in shaping our understanding and helping us successfully complete this work.

We extend our heartfelt thanks to our mentor, **Mr. Mairala Chinnarao**, for his expert guidance, practical insights, and consistent supervision at every stage of the project. His clarity in explaining concepts and willingness to assist us greatly contributed to the successful execution of this work.

We also express our sincere appreciation to the Head of the Department and all the faculty members of the Civil Engineering Department for their support and encouragement.

Finally, we would like to thank all our batchmates and team members for their cooperation, teamwork, and support, which helped us complete this project successfully.

TABLE OF CONTENTS

ABSTRACT

CHAPTER 1: INTRODUCTION

CHAPTER 2: OBJECTIVES

CHAPTER 3: LITERATURE REVIEW

CHAPTER 4: PROBLEM STATEMENT

CHAPTER 5: METHODOLOGY

CHAPTER 6: SYSTEM OVERVIEW

6.1 COMPONENT DESIGN

6.2 DESIGN AND MODELING

CHAPTER 7: WORKING PRINCIPLE

CHAPTER 8: RESULTS AND DISCUSSION

8.1 APPLICATIONS

8.2 ADVANTAGES

8.3 LIMITATIONS

8.4 FUTURE SCOPE

CHAPTER 9: CONCLUSION

ABSTRACT

The rapid advancement of modern infrastructure has created a growing demand for adaptive, efficient, and multifunctional structural systems. Conventional roofing systems used in large-span structures such as stadiums and auditoriums are often rigid, heavy, and incapable of adapting to environmental conditions. This project presents the design and development of an origami-based collapsible roof system that addresses these limitations through the application of folding geometry principles.

The proposed system utilizes the concept of origami engineering to transform a large surface area into a compact configuration and vice versa. The roof is designed as a dome-like structure supported by curved ribs and flexible sheet panels, which operate through a coordinated rotational mechanism. The integration of modern CAD tools such as SolidWorks enables precise modeling, assembly, and motion analysis of the structure.

This project demonstrates that origami-inspired designs can significantly enhance flexibility, reduce material consumption, and improve functionality in large-scale infrastructure. The system is particularly suitable for stadiums, temporary shelters, and adaptive architectural applications, offering both structural efficiency and aesthetic appeal.

CHAPTER 1: INTRODUCTION

Modern infrastructure is increasingly evolving toward systems that are not only structurally efficient but also adaptable to changing environmental and functional requirements. Large-span structures such as stadiums and exhibition halls demand roofing systems that can provide protection when required and openness when desired. Traditional roofing solutions, however, are often static in nature and fail to offer such flexibility.

Conventional roofs are typically heavy, expensive, and permanently fixed structures. These systems require significant material usage and are not capable of adjusting to varying weather conditions or functional demands. As a result, there is a need for innovative solutions that combine structural stability with dynamic adaptability.

Origami-based engineering provides a promising solution to this challenge. Origami, the traditional art of paper folding, has been successfully applied in various engineering domains, including aerospace structures, biomedical devices, and deployable systems. By incorporating folding geometry into structural design, it becomes possible to create systems that can expand and contract efficiently.

This project focuses on designing a collapsible dome roof system inspired by origami principles. The aim is to develop a structure that can be compactly folded and smoothly deployed while maintaining structural integrity and functionality.

CHAPTER 2: OBJECTIVES

The primary objective of this project is to design and develop a functional collapsible roof system using origami-based concepts. The system is intended to demonstrate the feasibility of folding structures in large-scale applications such as stadiums.

Another key objective is to develop accurate three-dimensional models using SolidWorks software. The modeling process involves creating individual components, assembling them, and analyzing their motion to ensure smooth operation.

The project also aims to achieve a reliable folding mechanism that ensures coordinated movement of all structural elements. Special attention is given to maintaining structural stability during both the opening and closing operations.

Furthermore, the design focuses on practical feasibility, ensuring that the proposed system can be implemented in real-world scenarios with appropriate material selection and manufacturing techniques.

Another important objective of this project is to explore the integration of mechanical systems with structural components. The interaction between moving parts such as hinges and rigid elements such as ribs requires careful design consideration to avoid failure during operation. Ensuring compatibility between these elements is essential for achieving long-term durability.

CHAPTER 3: LITERATURE REVIEW

The concept of deployable structures has been extensively studied in recent years, particularly in the fields of aerospace and civil engineering. Origami-inspired designs have gained attention due to their ability to transform shapes efficiently without compromising structural performance.

Researchers have explored various folding patterns such as Miura-ori and rigid foldable systems for applications in solar panels, satellite structures, and temporary shelters. These studies highlight the advantages of compact storage, lightweight construction, and rapid deployment.

In architectural engineering, retractable roofs have been implemented in stadiums; however, most existing systems rely on heavy mechanical components and complex mechanisms. These systems are often expensive and require high maintenance.

The integration of origami principles into structural design offers a more efficient alternative. By utilizing geometric folding techniques, it is possible to reduce mechanical complexity while enhancing adaptability and aesthetics.

CHAPTER 4: PROBLEM STATEMENT

Traditional roofing systems used in large-span structures suffer from several limitations. These systems are fixed in nature and cannot adapt to changing environmental conditions such as sunlight, rain, or wind.

Additionally, conventional roofs require large amounts of material, leading to increased construction costs and structural loads. The inability to modify or retract these roofs also results in inefficient space utilization.

Another major challenge is the complexity involved in designing movable roofing systems. Ensuring smooth motion, maintaining structural stability, and achieving lightweight construction are critical issues that need to be addressed.

Therefore, there is a need for a smart and flexible roofing system that can overcome these challenges while providing efficient performance and practical feasibility.

Another major issue associated with conventional roofing systems is their lack of versatility. Once constructed, these systems cannot be modified or adapted without significant reconstruction. This limits their usability in multi-purpose venues where flexibility is essential.

Environmental factors such as wind loads, temperature variations, and precipitation also pose challenges to fixed roof systems. In many cases, these structures are overdesigned to withstand worst-case scenarios, resulting in unnecessary material usage and increased costs. A collapsible system, on the other hand, can adapt to such conditions more efficiently.

CHAPTER 5: METHODOLOGY

The methodology adopted for this project involves a systematic approach starting from problem identification to final evaluation. Initially, the limitations of existing roofing systems were studied to understand the need for an adaptive solution.

The next stage involved studying various origami structures and identifying suitable folding patterns that can be applied to large-scale systems. Based on this study, a conceptual design was developed.

Two-dimensional sketches were created to visualize the structure and its components. These sketches were then converted into three-dimensional models using SolidWorks software. Each component was designed separately and later assembled to form the complete system.

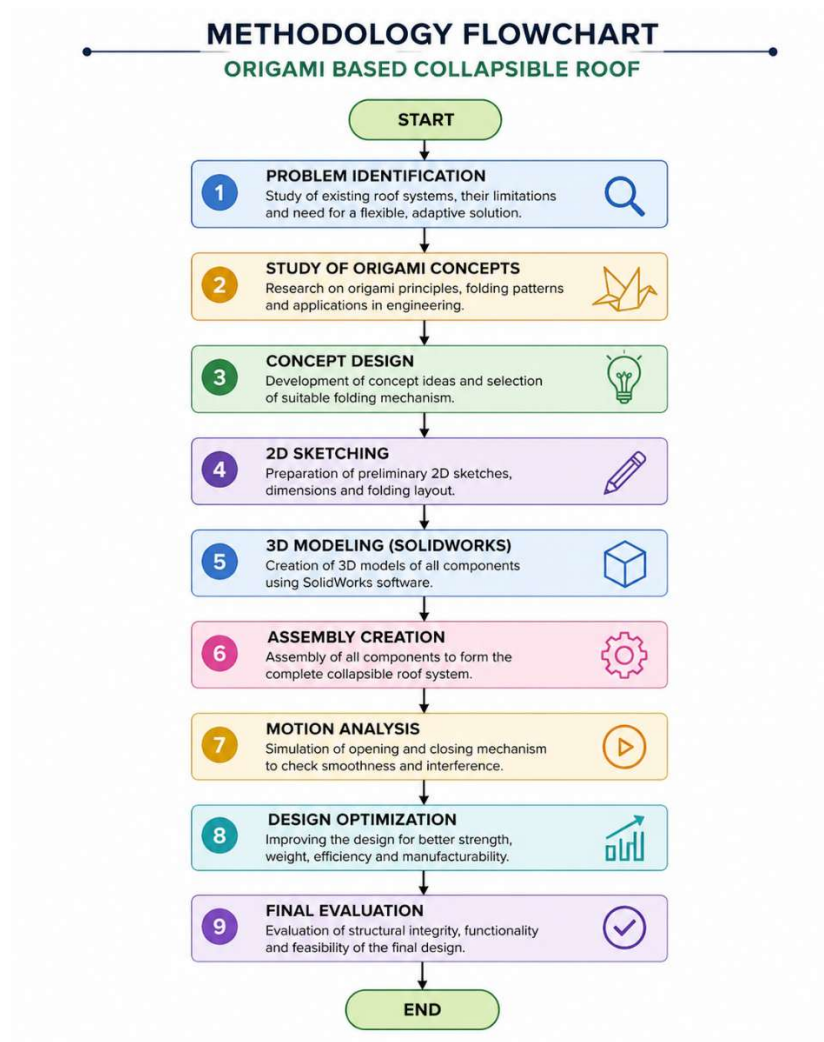
Motion analysis was performed to study the opening and closing mechanism of the roof. The final design was evaluated based on its structural performance, feasibility, and efficiency.

During the conceptual design phase, multiple configurations were evaluated to determine the most suitable folding mechanism. Factors such as ease of fabrication, structural stability, and motion efficiency were considered while selecting the final design. Iterative refinement played a key role in improving the system.

The modeling phase involved creating parametric designs that could be easily modified based on requirements. This approach allowed for quick adjustments in dimensions and geometry, enabling optimization of the structure. Each component was tested individually before being integrated into the assembly.

The motion analysis stage was particularly critical, as it helped identify potential issues such as collisions between components and uneven movement. By simulating real-world conditions, the design was refined to ensure smooth and synchronized operation.

Finally, the evaluation phase involved assessing the overall performance of the system in terms of functionality, feasibility, and efficiency. The results confirmed the viability of the proposed design.



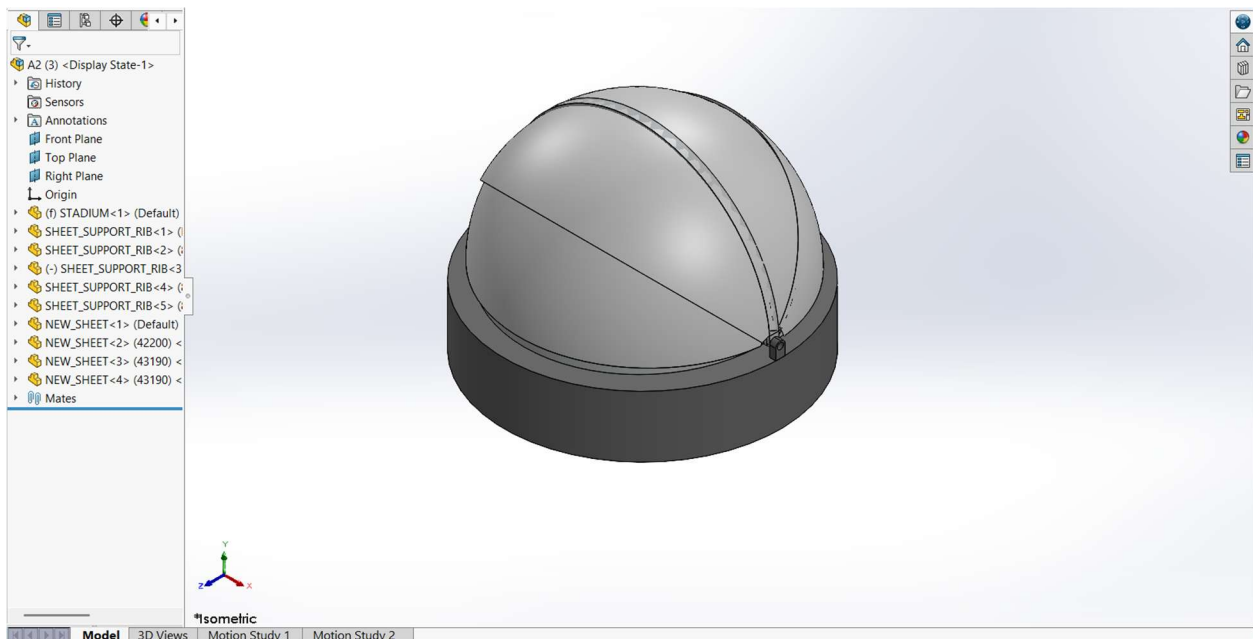
CHAPTER 6: SYSTEM OVERVIEW

The proposed system consists of a circular base structure that represents the stadium foundation. This base supports the entire roofing system and provides stability.

Curved ribs are attached to the base, forming the primary structural framework of the roof. These ribs are designed to guide the folding motion and maintain the dome shape.

Sheet panels are connected to the ribs and act as the covering elements. These panels are designed to fold along with the ribs during operation.

A central rotation mechanism is used to control the movement of the ribs and panels. This mechanism ensures synchronized motion, allowing the roof to open and close smoothly.



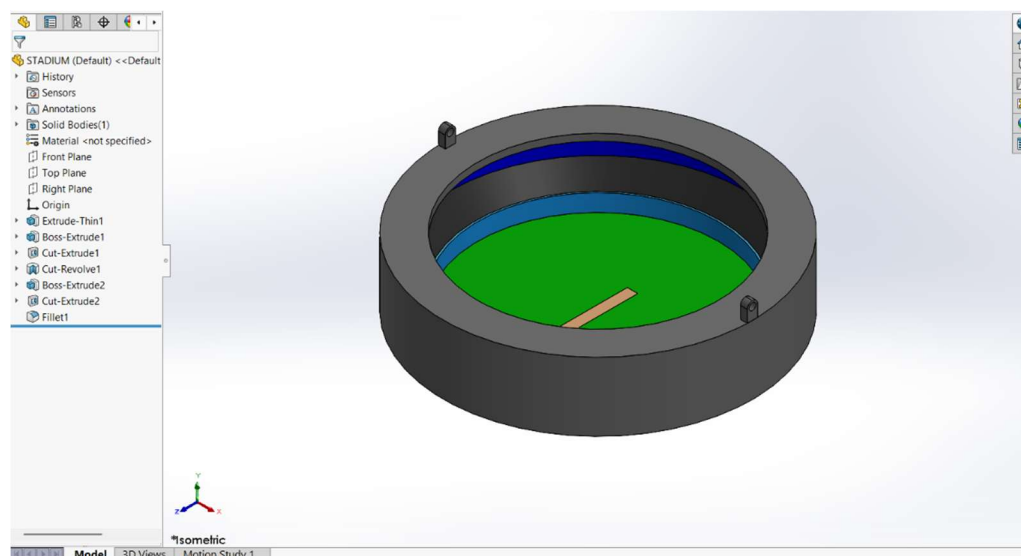
6.1 COMPONENT DESIGN

The base structure is designed with a circular geometry to distribute loads uniformly. It serves as the foundation for all other components and provides support to the ribs and hinges.

The rib design plays a crucial role in the system. Curved ribs of varying lengths are used to create the dome structure. These ribs not only support the sheet panels but also guide the folding motion.

The sheet panels are designed with a curved profile to match the ribs. The thickness of the panels is selected to ensure both flexibility and strength. These panels provide coverage and fold efficiently during operation.

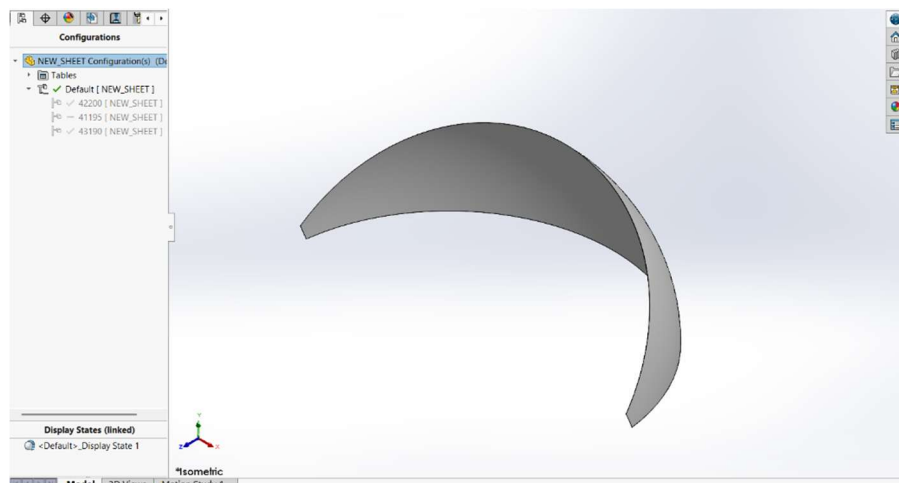
The central rotation mechanism is responsible for controlling the movement of the entire system. It ensures that all ribs move in coordination, resulting in smooth transformation.



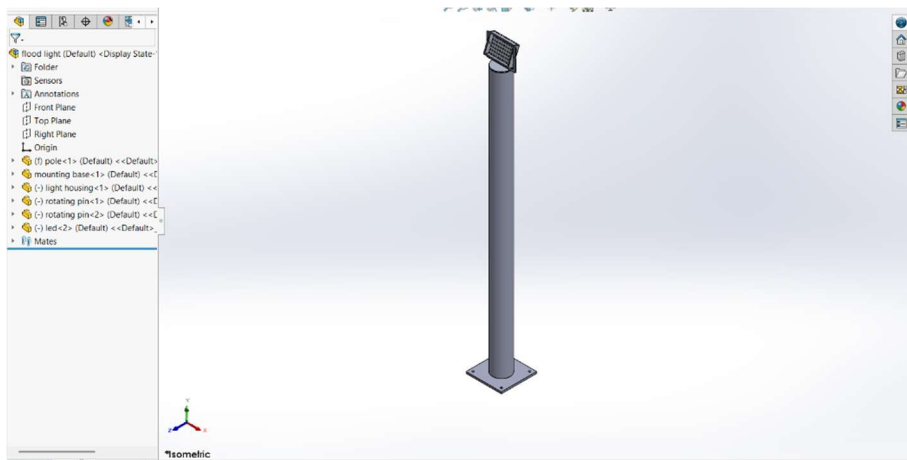
SUPPORT BASE



CURVED RIB



SHEET



FLOOD LIGHT SYSTEM

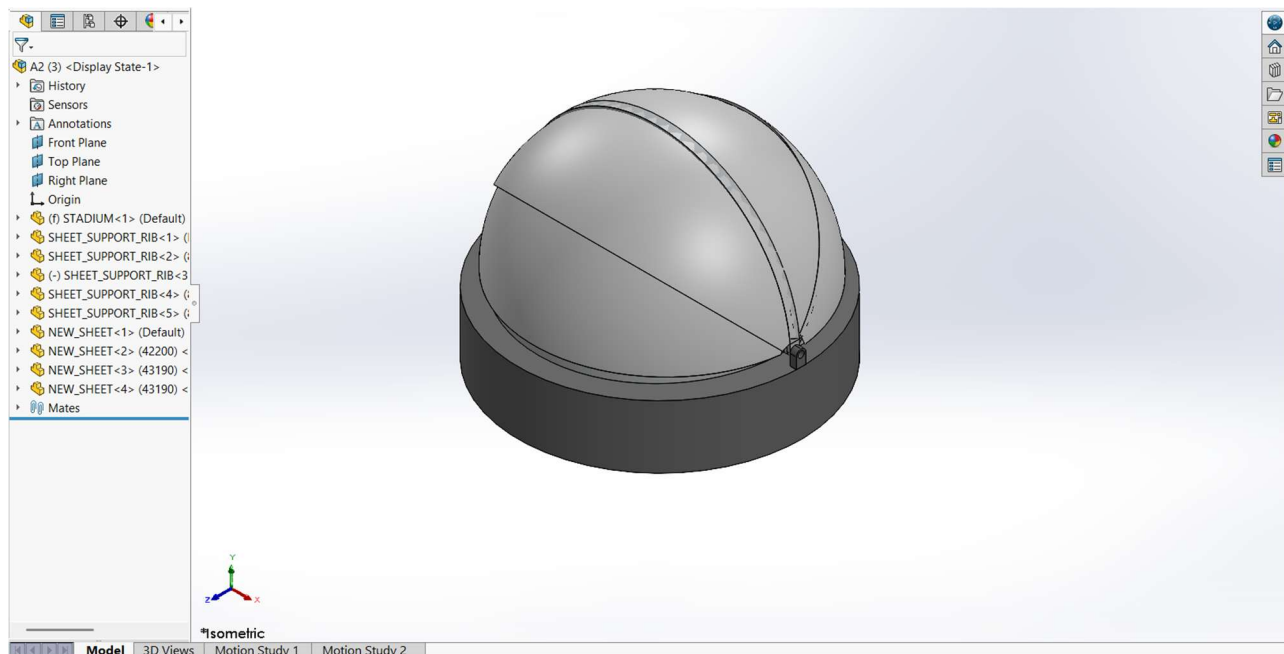
6.2 DESIGN AND MODELING

The design process was carried out using SolidWorks software. Each component was modeled individually with precise dimensions to ensure accuracy.

The base structure was created as a circular platform with provisions for mounting ribs and hinges. The ribs were designed with curved profiles and assembled radially.

Sheet panels were modeled to match the curvature of the ribs. These panels were connected using hinge joints to allow folding.

The complete assembly was created by combining all components. Motion simulation was performed to verify the functionality of the design and ensure smooth operation.



CHAPTER 7: WORKING PRINCIPLE

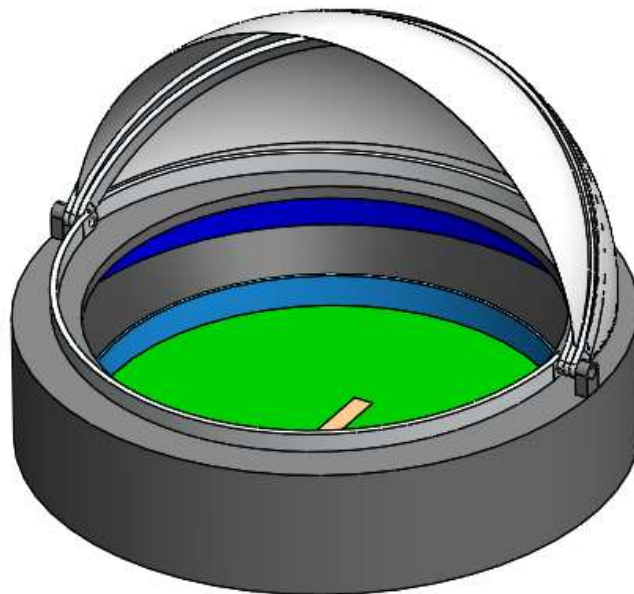
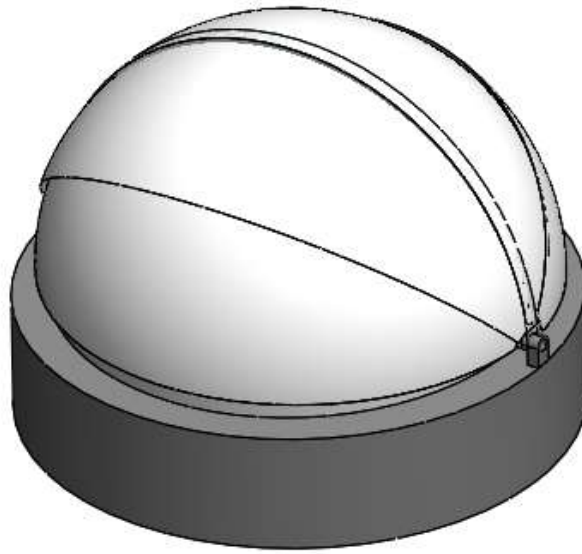
The working principle of the origami-based collapsible roof system is based on a coordinated rotational and folding mechanism that enables the structure to transform between open and closed configurations. The system primarily consists of a circular base structure, a series of curved ribs, flexible sheet panels, and a central rotation mechanism that controls the motion of the entire assembly. Each component is designed to function in synchronization to achieve smooth and efficient movement.

When the system is activated, the central rotation mechanism initiates motion, causing all the ribs to rotate simultaneously about their respective hinge or pivot connections. These ribs are arranged radially and are connected to both the base structure and the sheet panels. As the ribs begin to move outward, they guide the attached sheet panels to unfold, gradually expanding the roof structure into a dome-like covering. This process ensures that the entire surface area is covered uniformly without gaps or misalignment.

Conversely, when the rotation is reversed, the ribs move inward toward the center, and the sheet panels begin to fold along predefined lines. This folding action is inspired by origami principles, where large surfaces are transformed into compact forms through controlled geometric folding. The panels are designed with curvature and flexibility so that they can fold efficiently without causing structural interference or excessive stress.

The effectiveness of the system relies heavily on the synchronization of all moving parts. The hinge joints and connections are carefully designed to allow smooth rotation while maintaining structural stability. Any imbalance or delay in movement could lead to uneven loading or mechanical failure, so precise alignment and coordination are essential.

The mechanism can be operated manually or integrated with automated systems such as hydraulic or motor-driven controls. Automation enhances the ease of operation and allows for quicker response to environmental conditions. Additionally, the design minimizes resistance during motion, ensuring that the energy required for operation is relatively low.



CHAPTER 8:

8.1 APPLICATIONS

The proposed system has a wide range of applications in modern infrastructure. It can be used in sports stadiums to provide flexible roofing solutions.

It is also suitable for auditoriums and exhibition halls where adaptability is required. Temporary shelters and disaster relief structures can benefit from its portability and quick deployment.

In addition, the system has potential applications in military structures and space deployable systems, where compact storage and efficient deployment are essential.

8.2 ADVANTAGES

The origami-based collapsible roof offers several advantages over conventional systems. It provides a compact folding mechanism that allows efficient storage and transportation.

The structure is lightweight, reducing material usage and overall cost. The system is easy to operate and requires less mechanical complexity compared to traditional retractable roofs.

The design is space-efficient and enhances the aesthetic appeal of the structure. It represents a modern approach to architectural engineering.

8.3 LIMITATIONS

Despite its advantages, the system has certain limitations. The design of hinge joints is complex and requires high precision.

Manufacturing such structures demands advanced techniques and accurate fabrication. Maintenance of moving parts is also a concern, as wear and tear can affect performance.

The initial setup cost may be high due to the complexity of design and materials used.

8.4 FUTURE SCOPE

The project can be further enhanced by integrating hydraulic or automated systems to control the movement of the roof.

Solar panels can be incorporated into the sheet panels to generate renewable energy. The development of a scaled prototype can help in testing and improving the design.

Advanced materials can be used to improve durability and reduce weight, making the system more efficient and cost-effective.

CHAPTER 14: CONCLUSION

The project successfully demonstrates the design and development of an origami-based collapsible roof system. The use of folding geometry enables efficient transformation between compact and expanded states.

The system achieves smooth motion, structural stability, and practical feasibility. It provides a modern solution to the limitations of traditional roofing systems.

The proposed design has significant potential for application in adaptive infrastructure and represents a step forward in innovative structural engineering.